

THE FINITE DIFFERENCE TECHNIQUE FOR SOLVING CRACK PROBLEMS

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Abstract—A detailed examination of the Finite Difference method for solving crack problems is presented and discussed. The three classical mode I configurations (i.e. Centered Crack Plate, Double Edge Notch and Single Edge Notch) as well as an uncommon case (A Penny Shape Crack embedded in a circular plate in bending) are solved and discussed. The Stress Intensity Factors are computed by taking more than one (first) term in William's Series, using two or three points near the tip. This technique improves the accuracy and frees one from relying on the very first point near the tip as a measure base. In most cases, the accuracy was found to be between 1-3% for uniform mesh size in the order of 5% from the half crack length. No special imposed functions were used near the tip, which makes the technique competitive to the Finite Element method, especially for 3-D problems or cases where the degree of singularity is not known. The solution is found iteratively (a two step SOR method) and some techniques for quick convergence are discussed.

INTRODUCTION

MOST numerical solutions for crack problems have been done by the Finite Element Method (FEM)[1] with very little effort using the Finite Difference (F.D.) technique. The main reason is the geometrical flexibility of the FEM and the ability to impose a singularity function for the solution near the tip. However, for many 3-D problems, interlaminar cracks in composites and cases where the degree of singularity is not yet known, accurate numerical solutions are rare [2, 3]. Moreover, comparing numerical results for these cases with analytical solutions is very difficult, especially for finite bodies. Since 1973 [4] studies of crack problems using the F.D. method have been done continuously [5-6], for 2-D dynamic case, 3-D isotropic and orthotropic cases. Other singular problems, like stress concentrations near edges [7] and holes [8] in composites, showed the general capacity of the method. A basic study, to check the accuracy of the F.D. numerical solution around singular points is therefore needed to establish its validity and reliability.

THE DIFFERENCE EQUATIONS FOR ELASTIC SINGULAR PROBLEMS

The continuous elastic displacement field is transformed to distinct mesh points lying at specific locations in the field. A grid for bimaterial problems is chosen as a general example to demonstrate the numerical approximations for the displacement field and is shown in Fig. 1. The nodal points are divided into inner points and fictitious boundary points. Derivatives for inner points are computed by a regular central difference approximation, i.e.

$$U_x = [U(I+1, J) - U(I-1, J)]/(2\Delta x) \quad (1)$$

$$U_{xx} = [U(I+1, J) - 2U(I, J) + U(I-1, J)]/\Delta x^2 \quad (2)$$

$$U_{xy} = [U(I+1, J+1) + U(I-1, J-1) - U(I+1, J-1) - U(I-1, J+1)]/(4\Delta x\Delta y) \quad (3)$$

The other inner derivatives are derived in the same way with an accuracy $O(\Delta x^2)$. The accuracy can be higher (Δx^4) if one chooses more points near the location of the derivative.

In order to satisfy boundary conditions (b.c.) as well as fulfilling continuity, (as in bimaterial case, Fig. 1) a special artificial points are sited half mesh space beyond the borders of the real body. In this way, b.c. are imposed with the same degree of accuracy ($\Delta x^2, \Delta y^2$) as the internal equilibrium equations. Derivatives at the boundaries are obtained as follows (Fig. 1)

$$U_x(x, 0) = [U(I+1, 2) - U(I-1, 2)]/2\Delta x \quad (4)$$

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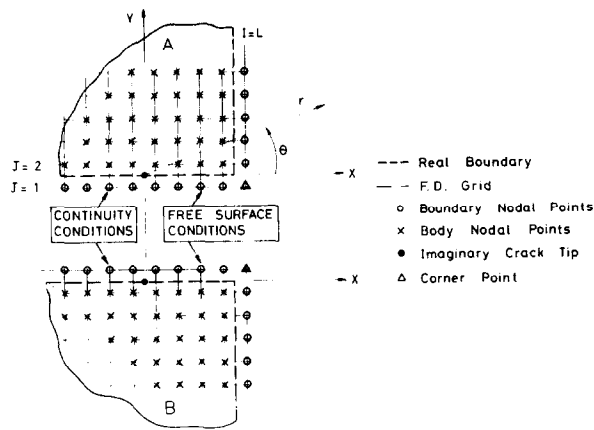


Fig. 1. The finite difference grid for dissimilar materials and/or cracks.

or

$$U_{,x}(x, 0) = [U(I + 1, 2) - U(I - 1, 2) + U(I + 1, 1) - U(I - 1, 1)]/4\Delta x. \tag{5}$$

The first option is less accurate but has the advantage of keeping the iterative process explicit. It is used where no considerable deflection of the boundary line (as in bending) is expected. In bending problems, especially with bimetals, one must use b.c. (5) to get accurate results. For boundaries of contact between two dissimilar materials there are two fictitious mesh points for every point of contact. The six displacements (3 for each point in the general 3-D case) are found by imposing the three equilibrium equations and three continuity relations. For the crack lips, each fictitious point has three free surface stress conditions.

The perpendicular derivative is simply

$$U_{,y}(x, 0) = [U(I, 2) - U(I, 1)]/\Delta y. \tag{6}$$

The location of the crack tip is assumed to be at the center of the mesh, as described in Fig. 1 and *never* on a mesh point. Results prove that this assumption gives accurate SIF evaluation even for large mesh sizes compared to the crack length. It is this feature which makes the F.D. technique so unique when dealing with crack problems. The corner points should be examined carefully. Since these points cannot fulfill all b.c. from all sides, they are computed by extrapolation from other adjacent boundary points. Experience shows that as long as the extrapolation does not cause divergency, results are not sensitive to what type of approximation is taken, especially for values far from the corner. The approximation taken here for the corner in material A (Fig. 1)

$$U(I = L, J = 1) = [2U(L - 1, 1) - U(L - 2, 1) + 2U(L, 2) - U(L, 3)]/2 \tag{7}$$

which is a first order averaged extrapolation. Other types of b.c. for the corner points which keep some of the conditions from either face have minor influence on the solution [7].

For the 2-D case, the stress equilibrium equations

$$\sigma_{ij,j} = 0 \quad i, j = 1, 2 \tag{8}$$

is transformed to the displacement equilibrium equations of the type

$$a_{11}U_{,xx} + a_{12}U_{,yy} + a_{13}V_{,xy} = 0 \tag{9a}$$

$$a_{21}V_{,xx} + a_{22}V_{,yy} + a_{23}U_{,xy} = 0 \tag{9b}$$

where U, V are the displacements in the x and y directions and a_{ij} are constants depending on the material modulus.

In the non isotropic case, or when the mesh is not rectangular [8] more derivatives of the displacement vector are present, and the coefficients a_{ij} are dependent on x and y . However, the F.D. method solves all cases by the same procedures.

CALCULATING THE STRESS INTENSITY FACTOR

Knowing the displacement field at all nodal points, the SIF can be calculated either from the near tip stresses along the crack direction or directly from the displacements of the crack lips near the tip [1]. In both methods, since no singularity is imposed on the numerical solution, the very near tip point (i.e. half mesh from the "imaginary" tip) carries the error of "smearing" the singularity. A typical example of this feature is shown in Fig. 2, where results for $\sigma_{yy}(r, \theta = 0)$ from a coarse mesh solution ($\Delta x = \Delta y = a/11$) were compared to those from a finer mesh ($\Delta x = \Delta y = a/33$). The two solutions had the major difference at the first mesh point near the tip (2.5%). All other points had less than 0.5% difference, and are more accurate. It also shows the consistency of the numerical scheme. Grid points, which are located far from the tip, cannot give the SIF results accurately, if the calculation is based on the first term in Williams' series [9]. Therefore, more than one term is taken here and a search for the optimum region and number of terms is conducted. For the line $\theta = 0$ (Fig. 1)

$$\sigma_{yy}(r) = b_1 r^{-1/2} + b_2 r^{1/2} + b_3 r^{3/2} + \dots \quad (10)$$

choosing n points one gets n equations from which b_i can be calculated directly. The matrix form of this calculation is

$$\begin{bmatrix} r_1^{-1/2} & r_1^{1/2} & r_1^{3/2} & \dots & r_1^{(2n-3)/2} \\ r_2^{-1/2} & r_2^{1/2} & r_2^{3/2} & \dots & r_2^{(2n-3)/2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ r_n^{-1/2} & r_n^{1/2} & r_n^{3/2} & \dots & r_n^{(2n-3)/2} \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} = \begin{bmatrix} \sigma_{yy}(r_1) \\ \sigma_{yy}(r_2) \\ \vdots \\ \sigma_{yy}(r_n) \end{bmatrix} \quad (11)$$

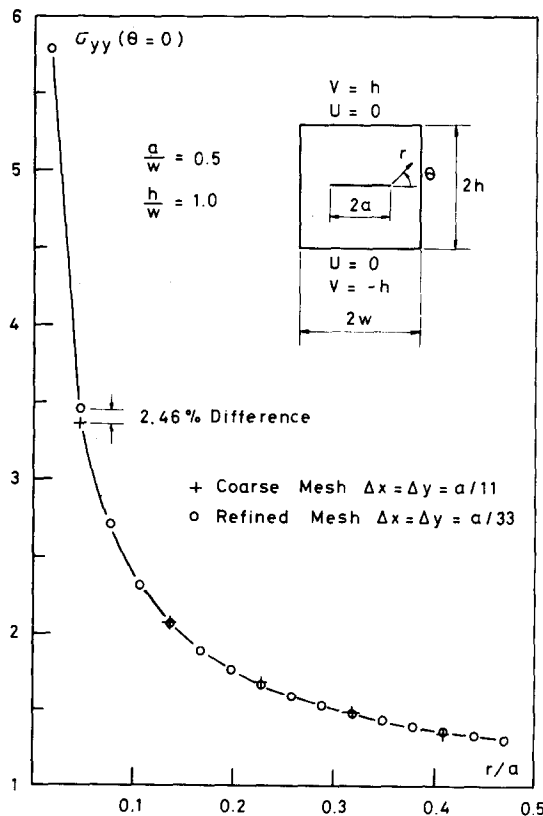


Fig. 2. The influence of mesh refinement on the stresses near the crack tip at the symmetry line ($\theta = 0$).

or

$$[R]\{B\} = \{\sigma_y\} \quad (12)$$

where b_1 is directly proportional to K_I .

The accuracy of K_I is influenced by two types of approximations. First, by neglecting all terms higher than n and second, by having numerical errors in $\sigma_{yy}(r_i)$. Assuming all r_i in $[R]$ bounded in a finite region $r < a$ (a is half crack length), when $n \rightarrow \infty$ $[R]$ becomes singular, causing infinite error in b_1 for finite error in $\sigma_{yy}(r_i)$. Therefore, the two approximations are balancing each other. For the near tip region, fewer terms are needed but the numerical accuracy of σ_{yy} is reduced, and for far-tip region, more terms are needed but the sensitivity of $b_1(K_I)$ to the accuracy of σ_{yy} is higher. To get a quantitative insight to the effect of n on the accuracy of $b_1(K_I)$, one can compare the first one and two terms. Taking the first term only ($\theta = 0$ through this analysis)

$$\sigma_{yy}(r_1) = b_1 r_1^{-1/2} \quad (13)$$

so that

$$\delta\sigma_{yy}(r_1) = \delta b_1 r_1^{-1/2} \quad (14)$$

where δ denotes the numerical error. Taking two first terms one gets

$$\delta b_1 = \left[\delta\sigma_{yy}(r_1)r_2^{1/2} - \delta\sigma_{yy}(r_2)r_1^{1/2} \right] / \left[\left(\frac{r_2}{r_1} \right)^{1/2} - \left(\frac{r_1}{r_2} \right)^{1/2} \right]. \quad (15)$$

where 1,2 are indices for the points chosen.

Since

$$\sigma_{yy} = d_1 U_{xx} + d_2 U_{yy} \quad (16)$$

where d_1 and d_2 are material constants and the leading error term of the central difference derivative is

$$\delta U_{xx} = \frac{1}{3} \Delta x^2 \frac{\partial^3 U}{\partial x^3}; \quad \delta V_{yy} = \frac{1}{3} \Delta y^2 \frac{\partial^3 V}{\partial y^3} \quad (17)$$

then, from (9), (15), (16) one gets

$$\delta\sigma_{yy}(r) = e_1 r^{-5/2} + e_2 r^{-3/2} + \dots \quad (18)$$

e_1, e_2 are constants, therefore

$$\delta\sigma_{yy}(r_1)/\delta\sigma_{yy}(r_2) = (e_1 r_1^{-5/2} + e_2 r_2^{-3/2} + \dots) / (e_1 r_2^{-5/2} + e_2 r_2^{-3/2} + \dots) \quad (19)$$

putting (19) in (15) and taking the first term one gets

$$\delta b_1 = \delta\sigma_{yy}(r_1) r_1^{1/2} (1 + k^{-1} + k^{-2}) \quad (20)$$

where $k = r_2/r_1$. Comparing (20) with (14) shows that the numerical accuracy of SIF calculated by taking two terms will be lower by a factor of 3 if the points will be taken close to each other, compared to their distance from the tip. If $k \rightarrow \infty$ one gets the best accuracy, which means that by "spreading" the points equally in the examined region, one will get the optimum results.

It is thus concluded, that for each case, in some region near the crack tip, very few terms (say, up to 3) will give accurate results and K_I will be less sensitive to $\delta\sigma_{yy}(r_i)$. Moreover, it is expected that by using more than one term, even coarse grid will give accurate results.

NUMERICAL EXAMPLES AND DISCUSSION

As a basis for comparison, the three classical mode I specimens have been examined. In Fig. 3, results for the Centered Crack Plate (CCP) type are compared to those from [10]. The accuracy (measured as the deviation per cent from the analytical solution) is shown for 1–4 terms of (10). For each

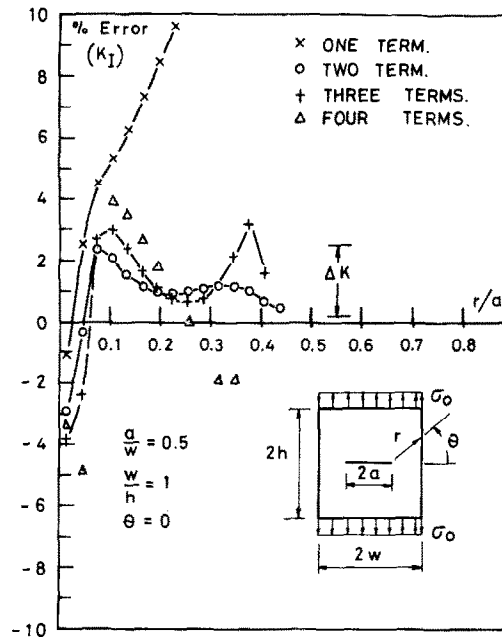


Fig. 3. The accuracy of the finite difference solution for the centered cracked plate as a function of the distance from the crack tip and number of points taken as data.

case, K_I is calculated at different distances from the crack tip. Each point in Fig. 3 sits at a distance from the smallest r_i taken. It is evident that the one term calculation cannot show a specific region for better accuracy. On the other hand, the calculation based on two and three terms (taken on the basis of adjacent points) shows a clear "flat" region with an accuracy of 1%. It is worth noting that most analytical solutions for finite bodies which are solved by series expansion of some kind have an error in the order of one per cent[11]. Moreover, this numerical accuracy is achieved at distances $r/a \approx 0.2$ and uniform mesh size $\Delta x/a \approx 1/35$ which is considered very coarse when using the Finite Element Method (without singularity imposed). It is also seen that the four terms calculation is less stable. An actual check of the stability of the solution can be achieved by changing one of the values $\sigma_{yy}(r_i)$ artificially by a quantity in the same order as the accuracy of the iterative process, and check its influence on K_I . In the

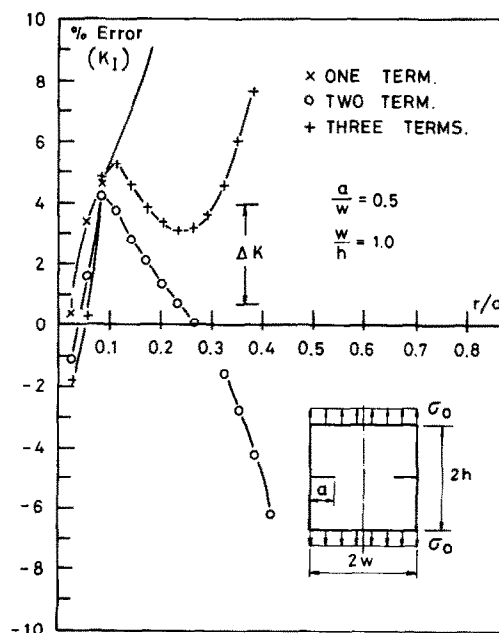


Fig. 4. The accuracy of the finite difference solution for the double edge notch specimen. ΔK -accuracy range.

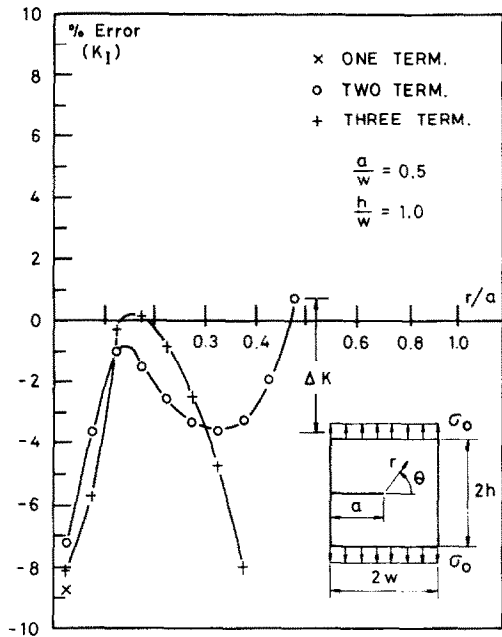


Fig. 5. The accuracy of the finite difference solution of the single edge notch specimen ($a/w = 0.5$).

“five terms calculation”, changing the fifth significant number in the stress values changed the error of K_I by $\sim 2\%$. For the case of two and three terms, no such sensitivity has been observed. These results confirm the analysis outlined above.

The same procedure for DEN and SEN specimen led to results shown in Figs. 4 and 5. Although the accuracy is somewhat reduced (3%), it is still in the range of their analytical solutions [12, 13]. A clear region, convenient for SIF calculation is seen and marked in the figures. Furthermore, the accuracy (ΔK in the figures) can be assessed by comparing the two and three terms calculation. For consistency and easy comparison in all three cases the geometrical relations $a/w = 0.5$ and $w/h = 1$ were taken. The SEN specimen is the most interesting case since it lacks the second symmetry (in the x direction). Since the solution is the displacement field, a displacement type of boundary condition at some point should be

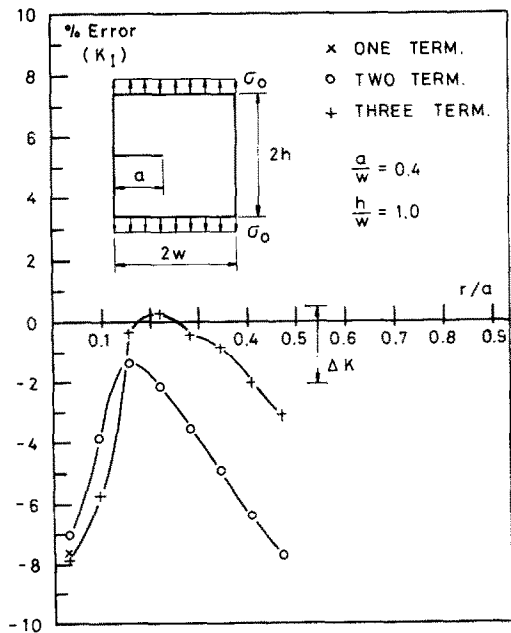


Fig. 6. The accuracy of the finite difference solution of the single edge notch specimen ($a/w = 0.4$).

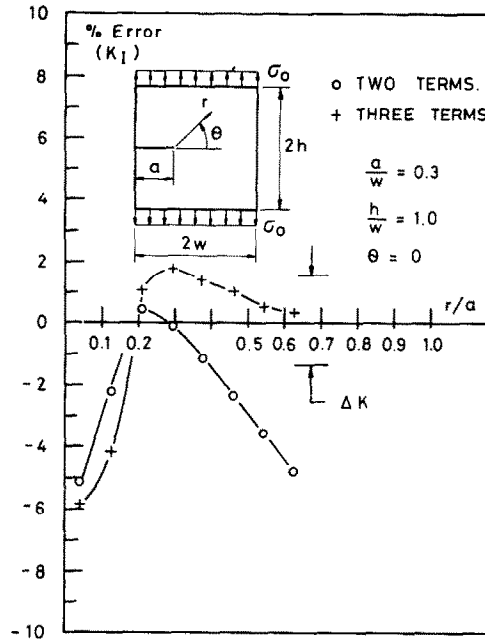


Fig. 7. The accuracy of the finite difference solution of the single edge notch specimen ($a/w = 0.3$).

imposed in order to cancel the rigid body motion from the solution. This disturbs in some way the numerical results since it violates either the equilibrium or the b.c. at this particular point. It is therefore necessary to check the SEN results carefully. First, two more crack lengths were solved ($a/w = 0.4, 0.3$) and results are shown in Figs. 6 and 7. It is seen that the accuracy is improved for smaller cracks (as in other F.E. solutions (1)) and the features of the numerical solution are the same. Furthermore, the mesh size for the $a/w = 0.3$ case is very coarse ($x/a \approx 1/12$) and the accuracy of K_I is still in the range of 2%. For all SEN cases, the additional condition was imposing the displacement in the x direction (U) at the point $r = (w - a)$, $\theta = 0$. To check the influence of this superfluous condition on the stress field, the normal stress σ_{yy} ($r, \theta = 0$) was plotted for two mesh sizes, with the finer one having twice nodal points

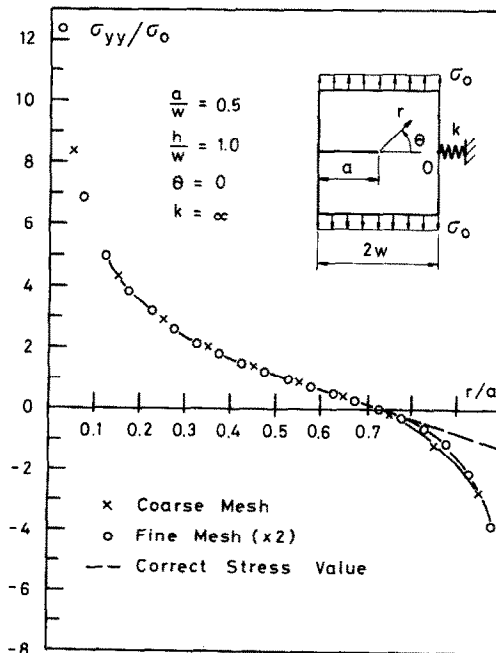


Fig. 8. The local effect of restraining U (point 0) on the stress field in the case of SEN specimen for two mesh grids.

in each direction (meaning that the accuracy is improved by a factor of 4) as seen in Fig. 8. It is concluded that the "rigid body motion condition" has a local influence on the stress field, and had no influence on the K_I calculation. However, this "influence check" is recommended for each case where superfluous conditions are imposed. Since the effective numerical iterative solution used here used multiple grids, this check comes as a by-product of the overall effort.

Another type of b.c. used is by adding an artificial spring to the specimen (Fig. 8). Results show that the displacement field is changed when changing the spring constant but the stress field remains the same. However, for absolute stiff spring ($k \rightarrow \infty$) the convergency is somewhat improved, so that "nailing" the point 0 was the best choice.

ON THE ITERATIVE SCHEME AND THE CONVERGENCE RATE

The equilibrium eqns (9) together with the proper boundary conditions are approximated by Finite Difference equations. The overall relation between the unknown displacement field and material constants can be put in the form

$$[A]\{u\} = \{b\} \quad (21)$$

where A is an $N \times N$ matrix, b a known $N \times 1$ vector and N^2 is the number of points in the field. Equation (21) is solved by an iterative semi (double step) SOR method. The main reasons for adopting iterative schemes is in its efficiency for solving very large systems as in 3-D problems [5]. For 2-D problems, the C.P.U. time is fair if a proper iterative technique is used. In each iteration, the "inner body" nodal points are corrected to impose local equilibrium, then the boundary points are corrected to impose boundary conditions. The overall iteration can be summarized as

$$u^{(n+1)} = Gu^{(n)} + k \quad n = 0, 1, 2, \quad (22)$$

where G is the iterative matrix and k depends on the boundary conditions. For rectangular uniform grid, G is independent of the coordinates. A necessary and sufficient condition for convergence is [14]

$$S(G) \equiv \max_{1 \leq i \leq N} |\lambda_i| < 1 \quad (23)$$

where λ_i is the i -th eigenvalue of the matrix G . An upper bound for $S(G)$ is found by the relation (Gershgorin theorem)

$$S(G) \leq \max_{i=1,2,\dots,N} \sum_{j=1}^N |g_{i,j}| \quad (24)$$

in the case of the equilibrium eqns (9), mixed derivatives (3) cause the sum of each line in G to be greater than 1, so that no conclusions can be made by applying eqn (24). Usually, the condition for convergency is dependent on the b.c. imposed, and the asymptotic rate of convergency is dependent on both b.c. and the mesh size [14]. A very efficient technique to speed up the convergence rate is based on starting the iterative process with a very coarse mesh (the minimal points in the field to keep the basic features of the problem). For this first stage, the convergence rate is higher and each iteration takes relatively very short time to execute. After converging, the coarse solution is used as a first approximation to a finer mesh and the process is repeated. A typical example is shown in Fig. 9. A comparison between a direct fine mesh and the Multiple Grid Technique (MGT) for the SEN specimen is shown. Here, ΔU max is the maximum difference between the $(n+1)$ and (n) displacement field, which is used as a convergency measurement. In most cases, ΔU max should reach the region $(0.1 \div 0.2) 10^{-4}$ to get an accurate solution when the displacement field is always normalised to give u max ≈ 1 . It is clearly seen that for a direct fine mesh the convergency rate is very slow or even zero. The MGT gives reasonable CPU time for the problems checked (~ 5 minutes or less on the 4341 IBM computer). It should be mentioned also, that the solution for SEN specimen took the longest time to converge and that CCP and DEN specimen converged after 2 minutes (average). A detailed analysis of the MGT method is studies in [15]. Although further reduction in the convergence time can be achieved (by multi-non uniform grids), the emphasis in this study was to get an accurate assessment of the SIF with a reasonable computer time rather than improving the efficiency of the program.

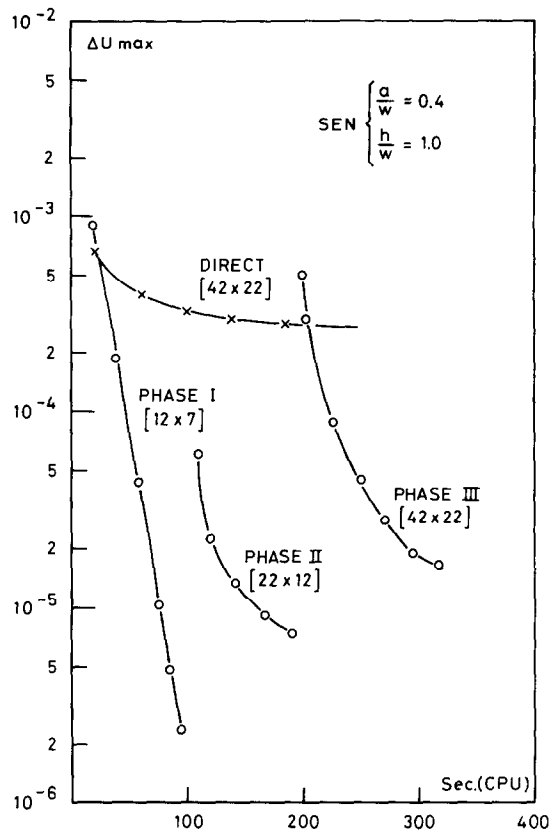


Fig. 9. The convergence rate for a direct fine mesh as compared to the Multigrid Method.

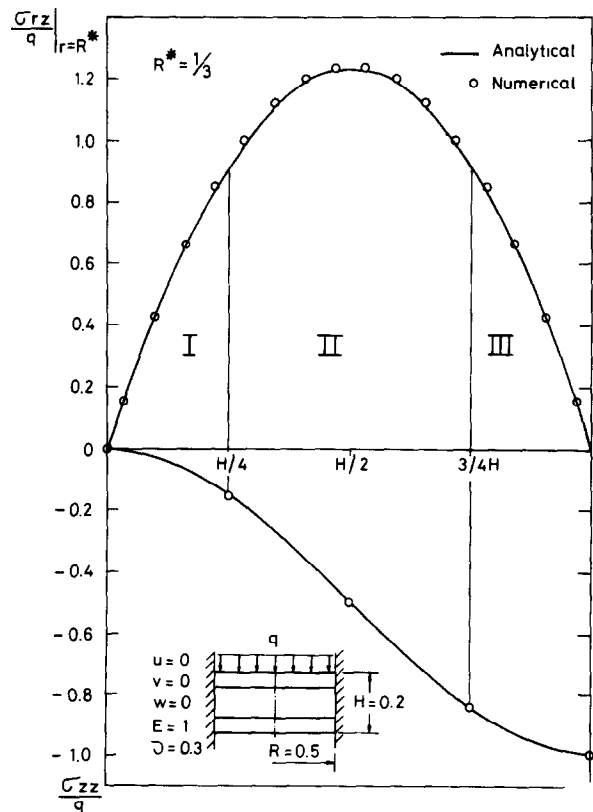


Fig. 10. Accuracy check of the finite difference solution for a circular clamped plate made of three identical isotropic layers.

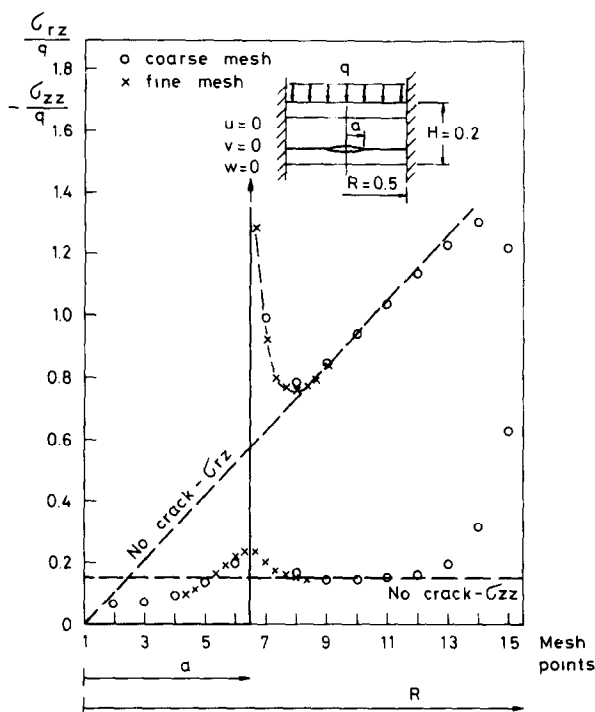


Fig. 11. Stress concentration around interlaminar crack for isotropic material.

APPLICATION FOR 3-D PROBLEMS

After establishing the validity of the F.D. technique for Classical 2-D problems, the method is used to find the interlaminar stress concentration around cracks in composites. Here, preliminary results concerning the accuracy and consistency of the numerical solution are discussed. In the general 3-D case, the 3 equilibrium equations are solved simultaneously together with the b.c. For circular plate (Fig. 10) these equations are solved in polar coordinates (r, θ, z) which is a particular case of the general coordinate transformation technique developed in [8]. For simplicity, a circular isotropic clamped plate is solved as a test case. Although all "three layers" are one in reality, the numerical procedure handles them like three different materials. The main reason for this was to check the influence of the boundary conditions between the layers on the numerical solution. The derivatives at the boundaries were taken as in eqn (5) since using eqn (4) led to erroneous results. Figure 10 shows the 3-D shear and normal stresses through the thickness at some radius $(r = R/2)$ as compared to analytical results [15] for a plate without any crack. Both stresses were found with less than 1% accuracy and as can be seen, the interlaminar boundary condition using eqn (5) did not cause any distortion in the interlaminar region.

A circular plate with a penny shaped crack is shown in Fig. 11. A coarse grid was applied first and after convergence, a three fold finer mesh was used to refine the field near the tip of the crack. Here, the MGT was applied to a specific region of the plate. Results from the two fields are shown as well as the stress values for "no crack" problem. The discrepancy near the clamped edge is due to imposing an "artificial" b.c. (i.e. $i = v = w = 0$) at this end.

Both K_I and K_{II} are present in this problem and while mode II is dominant and easier to calculate, the accuracy of mode I is not sufficient, while its influence on the crack growth may be considerable. A F.E. solution is needed here for accuracy check and comparison, which is a general necessity for all problems with no analytical solutions. The consistency of the F.D. solution is shown by the fine-to-coarse mesh comparison.

CONCLUSION

The following conclusions may be drawn from this study.

(1) The Finite Difference technique can be used successfully to solve crack problems using no special singular function.

(2) The method can be applied relatively easy to general 3-D crack problems, especially for layered media. However, other numerical methods should be used for the same problem as a comparative tool.

(3) Taking two or three points near the tip and using more than one term of Williams' series improves the SIF results and enables one to find K by relying on points which are not so close to the crack tip, where numerical errors are extreme.

(4) The efficiency of the iterative process can be improved by using a multigrid technique. Non uniform grids will probably improve the convergency further.

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